# Supplementary Materials: mtDNA-FM: A Domain-Specialized Foundation Model for Mitochondrial DNA

## S1. Hyperparameters

### S1.1 Pre-training

Table 1: Pre-training hyperparameters for Phase 1 and Phase 2.

| **Hyperparameter** | **Phase 1** | **Phase 2** |
|---|---|---|
| Training steps | 50,000 | 25,000 |
| Warmup steps | 2,000 | 500 |
| Peak learning rate | $1 \times 10^{-4}$ | $3 \times 10^{-5}$ |
| LR schedule | Cosine decay | Cosine decay |
| Batch size (effective) | 128 | 128 |
| Gradient accumulation steps | 8 | 8 |
| Optimizer | AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$) | AdamW |
| Weight decay | 0.01 | 0.01 |
| Gradient clipping | 1.0 | 1.0 |
| MLM masking probability | 0.15 | 0.15 |
| Heteroplasmy loss weight ($\lambda_{\text{het}}$) | 0.0 | 0.3 |
| Checkpoint rotation | Keep last 3 | Keep last 3 |
| Gradient checkpointing | Yes | Yes |

### S1.2 Model Architecture

### S1.3 Fine-tuning Hyperparameters

## S2. Mathematical Derivation: Circular Positional Encoding

### S2.1 Standard sinusoidal PE

In the original Transformer vaswani2017attention, position $p$ is encoded as:

$$\text{PE}(p, 2i) = \sin\left(\frac{p}{10000^{2i/d}}\right) \tag{1}$$

$$\text{PE}(p, 2i+1) = \cos\left(\frac{p}{10000^{2i/d}}\right) \tag{2}$$

For a sequence of length $L$, $\text{PE}(0, \cdot) \neq \text{PE}(L-1, \cdot)$ in general. As $d \to \infty$, the angle between PE(0) and PE($L-1$) grows monotonically with $L$.

Table 2: mtDNA-FM architecture hyperparameters.

| **Parameter** | **Value** |
|---|---|
| Hidden size ($d$) | 256 |
| Number of transformer layers | 6 |
| Number of attention heads | 8 |
| Feed-forward intermediate size | 1,024 |
| Attention head dimension | 32 |
| Dropout | 0.1 |
| Attention dropout | 0.1 |
| Max position embeddings | 514 (512 + [CLS] + [SEP]) |
| Vocabulary size | 4,102 |
| Activation function | GELU |
| Layer normalization $\epsilon$ | $10^{-12}$ |
| Total parameters | ≈5.8M |
| Positional encoding | Circular sinusoidal (fixed buffer) |
| Genome length for circular PE | 16,569 bp (rCRS) |

Table 3: LoRA fine-tuning hyperparameters for each downstream task.

| **Hyperparameter** | **Haplogroup** | **Pathogenicity** | **Heteroplasmy** |
|---|---|---|---|
| LoRA rank ($r$) | 8 | 4 | 4 |
| LoRA $\alpha$ | 16 | 8 | 8 |
| LoRA dropout | 0.05 | 0.05 | 0.05 |
| LoRA target modules | Q, V | Q, V | Q, V |
| Learning rate | $1 \times 10^{-3}$ | $5 \times 10^{-4}$ | $5 \times 10^{-4}$ |
| Epochs | 20 | 30 | 30 |
| Batch size | 32 | 16 | 16 |
| Loss function | Cross-entropy | BCE ($w_+ = 2.5$) | Huber ($\delta = 1.0$) |
| Weight decay | 0.0 | 0.1 | 0.1 |
| Input representation | CLS token | Variant position token | CLS token |

## S2.2 Circular modification

We replace $p$ with the circular angle $\phi_p = 2\pi p/L_{\text{genome}}$:

$$\text{PE}_{\text{circ}}(p, 2i) = \sin\left(\frac{2\pi p/L_{\text{genome}}}{10000^{2i/d}}\right) \tag{3}$$

$$\text{PE}_{\text{circ}}(p, 2i+1) = \cos\left(\frac{2\pi p/L_{\text{genome}}}{10000^{2i/d}}\right) \tag{4}$$

At $p = L_{\text{genome}}$:

$$\phi_{L_{\text{genome}}} = \frac{2\pi L_{\text{genome}}}{L_{\text{genome}}} = 2\pi$$

and $\sin(2\pi) = \sin(0) = 0$, $\cos(2\pi) = \cos(0) = 1$.

Therefore $\text{PE}_{\text{circ}}(L_{\text{genome}}, \cdot) = \text{PE}_{\text{circ}}(0, \cdot)$, which is the desired circular continuity property.

## S2.3 Properties

- *Uniqueness:* Positions $p$ and $p'$ with $|p - p'| < L_{\text{genome}}/2$ receive distinct encodings (the mapping $p \mapsto \phi_p$ is injective on $[0, L_{\text{genome}})$).

- *Smoothness at junction:* $\|\mathrm{PE}_{\mathrm{circ}}(p) - \mathrm{PE}_{\mathrm{circ}}(p+1)\|$ is constant across all positions including $p = L_{\mathrm{genome}} - 1$ (the circular junction), whereas for standard sinusoidal PE this distance jumps at position 0.
- *Non-learnable:* Implemented as `nn.Buffer`, not `nn.Parameter`; gradients do not flow through the positional encoding.

# S3. Per-class Haplogroup Metrics

Results from LoRA fine-tuning under CPU compute constraints (2 epochs; class collapse to haplogroups D, G, R). Non-zero recall for D (24.4%), G (62.3%), and R (13.6%) reflects the partial model: these three haplogroups have the largest training window counts and were the only predicted classes. All other classes have zero recall due to class collapse. See Section 4.2 of the main text for full discussion.

Table 4: Per-class haplogroup classification metrics from LoRA fine-tuning (CPU, 2 epochs). $N$ test refers to the number of test windows (not sequences) per class; each genome contributes ≈65 windows.

| **Haplogroup** | **Precision** | **Recall** | **F1** | **N windows** |
|---|---|---|---|---|
| A | 0.000 | 0.000 | 0.000 | 3,250 |
| B | 0.000 | 0.000 | 0.000 | 7,215 |
| C | 0.000 | 0.000 | 0.000 | 4,160 |
| D | 0.040 | 0.244 | 0.069 | 2,925 |
| E | 0.000 | 0.000 | 0.000 | 390 |
| F | 0.000 | 0.000 | 0.000 | 2,145 |
| G | 0.011 | 0.623 | 0.021 | 780 |
| H | 0.000 | 0.000 | 0.000 | 9,945 |
| HV | 0.000 | 0.000 | 0.000 | 845 |
| I | 0.000 | 0.000 | 0.000 | 1,170 |
| J | 0.000 | 0.000 | 0.000 | 2,470 |
| K | 0.000 | 0.000 | 0.000 | 1,300 |
| L0 | 0.000 | 0.000 | 0.000 | 3,575 |
| L1 | 0.000 | 0.000 | 0.000 | 1,820 |
| L2 | 0.000 | 0.000 | 0.000 | 4,485 |
| L3 | 0.000 | 0.000 | 0.000 | 5,265 |
| L4 | 0.000 | 0.000 | 0.000 | 780 |
| L5 | 0.000 | 0.000 | 0.000 | 260 |
| M | 0.000 | 0.000 | 0.000 | 5,265 |
| N | 0.000 | 0.000 | 0.000 | 1,170 |
| R | 0.015 | 0.136 | 0.026 | 1,040 |
| T | 0.000 | 0.000 | 0.000 | 4,160 |
| U | 0.000 | 0.000 | 0.000 | 6,760 |
| V | 0.000 | 0.000 | 0.000 | 975 |
| W | 0.000 | 0.000 | 0.000 | 520 |
| X | 0.000 | 0.000 | 0.000 | 585 |
| **Macro avg** | 0.003 | 0.039 | 0.004 | 73,255 |

# S4. Dataset Summary

# S5. DVC Pipeline

The complete analysis pipeline is managed with DVC. The following stages are defined:

Table 5: Datasets used in this study.

| Dataset | Source | Size | Purpose |
|---|---|---|---|
| HmtDB | clima2017hmtdb | 34,975 seqs used (47,000 total in database) | Phase 2 pre-train, fine-t |
| NCBI vertebrate mtDNA | NCBI Entrez | 117,615 seqs | Phase 1 pre-train |
| PhyloTree | vanoven2009phylotree | 26 major groups | Haplogroup labels |
| gnomAD v3.1 chrM | karczewski2020gnomad | ∼5,000 vars | Benign variants |
| ClinVar | landrum2018clinvar | ∼2,000 vars | Pathogenic variants |
| MITOMAP | lott2013mitomap | ∼X vars | External validation |
| HelixMTdb | bolze2019helixmtdb | ∼X vars | External validation |

1. `download_hmtdb`: Download and cache HmtDB sequences
2. `download_ncbi`: Download vertebrate mtDNA from NCBI Entrez
3. `download_variants`: Download gnomAD, ClinVar, PhyloTree variant data
4. `preprocess`: Quality filter, normalize to rCRS length, train/val/test split
5. `build_vocabulary`: Construct 4,102-token k-mer vocabulary
6. `pretrain_phase1`: Phase 1 cross-species pre-training (50k steps)
7. `pretrain_phase2`: Phase 2 human specialization (25k steps)
8. `finetune_haplogroup`: LoRA fine-tuning for haplogroup classification
9. `finetune_pathogenicity`: LoRA fine-tuning for pathogenicity prediction
10. `finetune_heteroplasmy`: LoRA fine-tuning for heteroplasmy regression
11. `evaluate`: Compute all metrics, generate evaluation reports

Reproduce the full pipeline: `dvc repro`

## S6. Ablation Learning Curves

Ablation training curves (circular PE vs. sinusoidal PE vs. learnable PE; two-phase vs. single-phase training; with vs. without heteroplasmy channel) will be included in the next version of this preprint following GPU-accelerated ablation runs.

## References